

# IOT - Semester - 9 - Vraj Suratwala

## Demo 1 : IR Sensor

Understanding The Configuration of IR Sensor.

IR Sensor is having main three pins.

VCC - (3.3v or 5v)

GND - Ground Pin

OUTPUT PIN - High and Low Voltage Indicators

## Demo 1.1 Normal IR Sensor Communication with Raspberry Pi

Problem Statement :

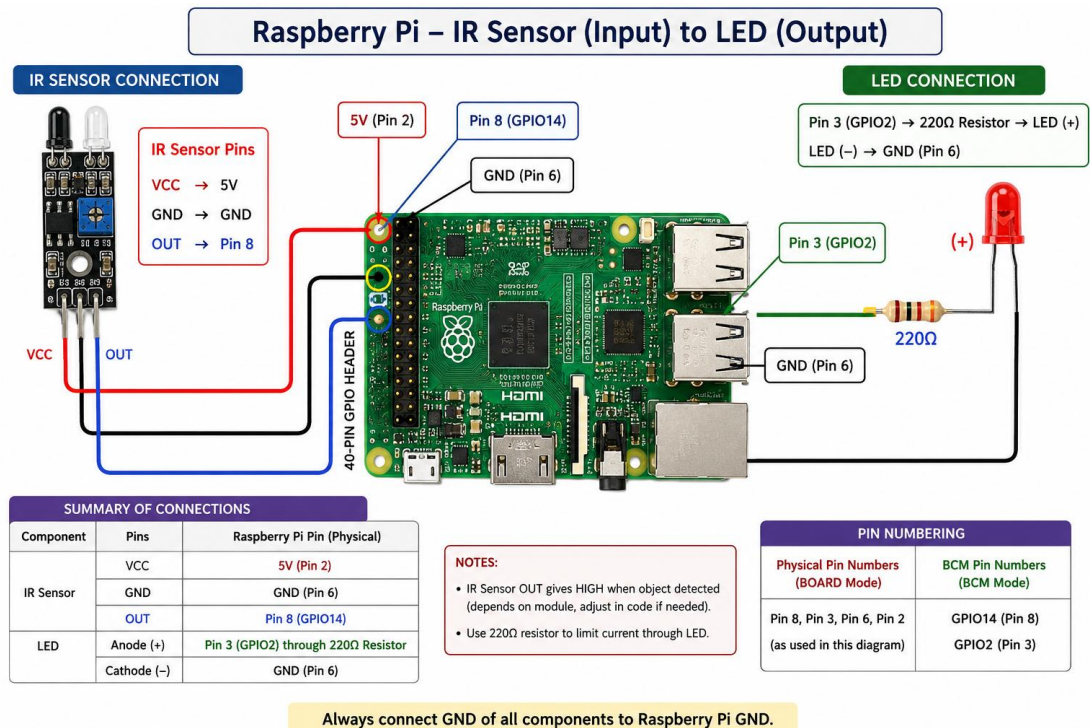
Write an IoT Program which will use a IR Sensor to Detect the Obstacle in the room and Send data to MCU (which will be Raspberry Pi) . MCU will process the data and blink an external LED.

Component Required :

- Raspberry Pi
- IR Sensor
- LED
- 220 Ohm. Register
- Breadboard
- Jumper Wires

Language Config : Python

Understanding The Connection Via Diagrams :



Code : In Python

```
import RPi.GPIO as IO

IO.setwarnings(False)
IO.setmode(IO.Mode)
//IO.setmode(IO.BCM)

Input_Data_Pin = 8
Output_Data_Pin = 3

IO.setup(Input_Data_Pin, IO.IN)
IO.setup(Output_Data_Pin, IO.OUT)

While 1:
    If(IO.input(8) == True):

        print("Obstacle")
        IO.output(Output_Data_Pin, True)
    else:
        print("Not Detected")
        IO.output(Output_Data_Pin, False)
```

## Demo 1.2 Normal IR Sensor - Raspberry Pi Serial Communication with Arduino Uno.

### Problem Statement :

Write an IOT Program which takes data where Raspberry Pi is Receiving the data from IR Sensor and Transmit that data to Arduino UNO.

Note : while Arduino UNO is Receiving the Data that time an internal LED should be blink.

### Component Required :

- Raspberry Pi
- IR Sensor
- 220 ohm. Register
- Breadboard
- Arduino UNO
- Jumper Wires.

Code : Python for Raspberry Pi

```
import RPi.GPIO as IO
import serial
import time
```

```
IO.setmode(IO.BCM)
IO_PIN = 14
IO.setmode(IR_PIN, IO.IN)
```

```
Arduino_Data_Variable = serial.Serial('/dev/ttyACMO', 9600)
```

```
Time.sleep(2)
```

```
If IO.input(TR_PIN) :
    print("Object Detected")
    Arduino_Data_Variable.write(b'1')
```

```
else:  
    print("No Object")  
    Arduino_Data_Variable.write(b'0')
```

```
Time.sleep(0.1)
```

---

Arduino Code : C++

```
Char data;
```

```
void setup(){
```

```
    Serial.begin(9600);
```

```
    pinMode(13, OUTPUT);
```

```
}
```

```
void loop(){
```

```
    If(serial.available()){
```

```
        Data = Serial.read();
```

```
        If(data == '1')
```

```
        {
```

```
            digitalWrite(13,High);
```

```
        }
```

```
        If(data == '0')
```

```
        {
```

```
            digitalWrite(13, Low);
```

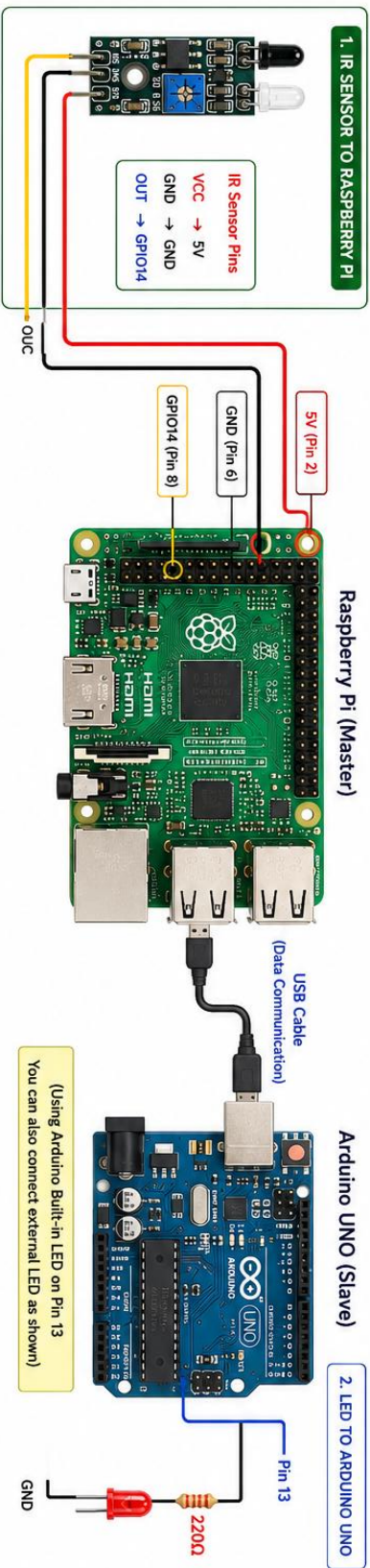
```
        }
```

```
    }
```

```
}
```

Let's Understand The Connections as well!

## Raspberry Pi (Sensor Input) → Arduino UNO → LED Blink



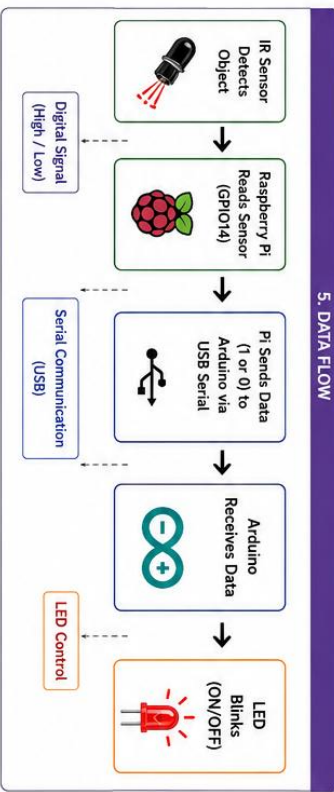
### 3. RASPBERRY PI CODE (Python)

```
1 import RPi.GPIO as GPIO
2 import serial
3 import time
4
5 GPIO.setmode(GPIO.BCM)
6 IR_PIN = 14 # GPIO14 connected to IR sensor OUT
7 GPIO.setup(IR_PIN, GPIO.IN)
8
9 # Open serial connection to Arduino
10 arduino = serial.Serial('/dev/ttyACM0', 9600)
11 time.sleep(2) # Give some time for connection
12
13 try:
14     while True:
15         if GPIO.input(IR_PIN):
16             print("Object Detected")
17             arduino.write(b'1') # Send 1 to Arduino
18         else:
19             print("No Object")
20             arduino.write(b'0') # Send 0 to Arduino
21             time.sleep(0.1)
22 except KeyboardInterrupt:
23     GPIO.cleanup()
24 arduino.close()
25
```

### 4. ARDUINO CODE (C++)

```
1 char data;
2
3 void setup()
4 {
5     Serial.begin(9600); // Built-in LED
6     pinMode(13, OUTPUT);
7 }
8
9 void loop()
10 {
11     if (Serial.available())
12     {
13         data = Serial.read();
14
15         if (data == '1')
16         {
17             digitalWrite(13, HIGH); // LED ON
18         }
19         if (data == '0')
20         {
21             digitalWrite(13, LOW); // LED OFF
22         }
23     }
24 }
```

### 5. DATA FLOW



### 6. IMPORTANT COMMANDS (Raspberry Pi)

- Install serial library  
`pip3 install pyserial`
- Check Arduino port  
`ls /dev/tty*`  
(Usually `/dev/ttyACM0` or `/dev/ttyUSB0`)

### 7. SUMMARY

- IR Sensor data → Raspberry Pi
- Raspberry Pi processes and sends data to Arduino via USB Serial
- Arduino receives data and controls LED
- This architecture is useful for advanced IoT / Robotics / AI projects

Additional Advanced Task to Perform :

1. Send The Text Data Rather then 0 & 1 in Serial Communication Data Flow.

2. Rather then using an internal LED, Use External LED to Blink with the time of 3s for 3 times.

3. For More Complex Level : Receive the data from Arduino UNO and Pass it to Raspberry PI Again using Duplex Communication & and Blink External LED Again.

4. Receive IR Sensor Data from Arduino and and Send Raspberry PI and Blink a LED.

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Thank You,  
Vraj Suratwala

Reference :  
LLM Models